ASSIGNMENT 1.1

Project Ideation, Environment Modeling, and PEAS Matrix Definition

AGENT TECHNOLOGY • 21CSE525T

Proposed Agent	Academic Deadline & Study Planning Agent (ADSPA)
Agent Persona	A personal academic planning assistant that monitors deadlines, prioritizes work, and adapts a study schedule when circumstances change.
Core Domain	Student task management, deadline awareness, and adaptive study planning.
Primary Objective	Help a student complete important academic tasks on time while balancing workload, available study hours, and task difficulty.

Design specification prepared to address the Assignment 1.1 deliverables: domain definition, PEAS profile, environment classification, and a structural comparison with a standard object-oriented program.

Student Name: _____

Register Number: _____

Date: _____

1. Domain Definition and Problem Statement

1.1 Domain

The Academic Deadline & Study Planning Agent (ADSPA) operates in the domain of academic task planning. Students typically manage assignments, quizzes, examinations, project milestones, and personal study goals at the same time. Deadlines may overlap, task duration can be uncertain, and unexpected events can reduce the time available. A static timetable may therefore become outdated quickly.

1.2 Problem Statement

Design an autonomous agent that collects academic task information and the student's available study time, estimates task urgency and workload, creates a prioritized schedule, and revises that schedule when new tasks, deadline changes, task progress, or time constraints are detected. The agent should recommend or schedule actions, explain its priorities, and ask for confirmation before making consequential changes such as deleting a task or sending a message.

1.3 Agent Persona

Name: ADSPA — Academic Deadline & Study Planning Agent
Role: A proactive digital study coordinator.
Operating style: Goal-directed and adaptive. It does not merely wait for a user to request a timetable; it checks task state, detects conflicts, and proposes revisions when relevant conditions change.

1.4 Goals and Scope

Goal	Description
Deadline reliability	Reduce missed deadlines by identifying urgent and high-impact tasks early.
Workload balance	Distribute work across available study periods instead of overloading a single day.
Adaptive planning	Recalculate the schedule when tasks, deadlines, progress, or availability change.
User control	Show the reasoning behind priorities and request approval for important external actions.

Out of scope: The first version will not submit assignments, take examinations, or make academic decisions on behalf of the student. It supports planning and reminders; the student remains responsible for the work.

2. PEAS Profile

PEAS stands for Performance measure, Environment, Actuators, and Sensors. It specifies how the agent's success is evaluated, where it operates, how it can act, and what information it can observe.

PEAS Element	Definition for ADSPA	Examples / Details
Performance Measure	Measures how well the agent achieves its planning goals.	• Percentage of tasks completed before deadlines • Number of missed deadlines • Schedule feasibility • Balance of daily workload • Reminder accuracy and timeliness • User acceptance of proposed changes
Environment	The digital and personal context in which planning takes place.	• Student's task list and calendar • Assignment and examination deadlines • Estimated task durations and difficulty • Available study hours • Calendar events and breaks • Notification service and device • Changing instructor requirements or newly assigned work

PEAS Element	Definition for ADSPA	Examples / Details
Actuators	Mechanisms through which the agent changes or communicates with the environment.	• Create or revise a study schedule • Rank tasks by priority • Display recommendations • Send reminders or alerts • Add approved study blocks to a calendar • Request confirmation when a conflict requires user input
Sensors	Inputs used to perceive task state and student constraints.	• User-entered tasks and deadlines • Calendar events and free/busy periods • Task completion check-ins • Time remaining before deadlines • User-defined priority and difficulty • Updates to tasks or deadlines • Feedback such as “too much work today”

3. Environment Characteristics Classification

The agent's environment can be classified using common dimensions of intelligent-agent environments. These classifications guide the design of its reasoning and planning process.

Characteristic	Classification	Justification
Observability	**Partially observable**	The agent can read recorded tasks and calendar events, but cannot directly know the student's true concentration, fatigue, or exact time needed unless the student reports it.
Determinism	**Stochastic**	Task duration, interruptions, unexpected work, and changes in personal availability are uncertain. The same schedule may not produce the same outcome every time.
Episodic vs. sequential	**Sequential**	A scheduling decision affects later decisions. Allocating time to one task changes the time available for other tasks.
Static vs. dynamic	**Dynamic**	Deadlines, task status, calendar events, and student availability can change while the agent is operating.
Discrete vs. continuous	**Mostly discrete, with continuous time**	Tasks and calendar slots are discrete objects, while time remaining and task duration are measured on a continuous scale.
Single-agent vs. multi-agent	**Primarily single-agent**	One planning agent makes recommendations for one student. External services, such as a calendar or notification service, are tools rather than competing planning agents in the initial design.
Known vs. unknown	**Partly known**	The agent's basic scheduling rules are known, but estimates of task duration and how the student responds to reminders may be uncertain and should improve from feedback.

3.1 Example of Autonomous Operation

Suppose the student has an assignment due tomorrow, an examination in three days, and only two free hours tonight. The agent senses the deadlines and calendar availability, evaluates urgency and estimated effort, and proposes a feasible plan. If a new class event later removes one study hour, the agent detects the conflict and recalculates the plan. It then explains the change and asks the student to approve any calendar edits. This observe–evaluate–act–reassess loop is the central autonomous behavior.

3.2 Illustrative Priority Logic

A simple first version can calculate a priority score from deadline urgency, task importance, and estimated effort. For example: **Priority Score = 0.5 × Urgency + 0.3 × Importance + 0.2 × Difficulty**, where each input is normalized to a 0–10 scale. The score is an illustrative design choice, not a universal formula. The agent should also check whether a task can fit into the available time before placing it in the schedule.

4. Autonomous Agent vs. Standard Object-Oriented Program

An autonomous agent may be implemented using object-oriented programming, but the two concepts are not identical. Object-oriented programming organizes software around objects, data, and methods. An autonomous agent is distinguished by its ongoing perception of an environment, goal-directed decision-making, and selection of actions based on observed conditions.

Dimension	Autonomous ADSPA Agent	Standard Object-Oriented Program
Initiation	Can periodically monitor tasks and detect a change without waiting for a new user command.	Typically executes a method after a user or another component calls it.
Inputs	Continuously or periodically gathers task, calendar, progress, and availability information.	Usually processes the data explicitly supplied to a particular function or method.

Dimension	Autonomous ADSPA Agent	Standard Object-Oriented Program
Decision logic	Chooses among possible actions according to goals, constraints, and current observations.	Follows the logic encoded in the called methods; it may not independently choose goals or actions.
Adaptation	Replans when deadlines, progress, or available time changes.	Requires an explicit call or programmed event handler to run the relevant update logic.
Control flow	Uses a recurring observe–reason–act loop, with event triggers as appropriate.	Often follows a predefined sequence of calls and control flow.
Goal orientation	Evaluates actions by outcomes such as deadline completion and schedule feasibility.	Performs the specific operations for which its methods were designed.
Example	Notices a deadline conflict, rebuilds the schedule, and alerts the student.	Creates a timetable only after the student clicks “Generate Schedule.”

5. Proposed High-Level Operating Cycle

Step	Agent Activity	Output
1. Perceive	Read task list, deadlines, calendar availability, and task progress.	Current state of the student's workload.
2. Interpret	Identify urgent tasks, conflicts, incomplete tasks, and missing information.	A set of constraints and planning priorities.
3. Decide	Score candidate tasks and choose a feasible order and study slots.	A proposed study plan.
4. Act	Show the plan, issue reminders, or make approved calendar updates.	Updated schedule and user notification.
5. Reassess	Check for new events, changed deadlines, and task completion.	Continue with the plan or trigger replanning.

6. Conclusion

The Academic Deadline & Study Planning Agent is a goal-directed system designed to help students manage academic work under changing constraints. Its PEAS profile defines measurable success, the operating environment, available actions, and information sources. The environment is partially observable, stochastic, sequential, and dynamic, so the agent should reassess conditions and revise its plan instead of relying on a fixed timetable. Compared with a conventional object-oriented program that responds to explicit calls, the proposed agent is designed around autonomous monitoring, goal-based decision-making, and adaptive action.

7. References

[1] 21CSE525T: Agent Technology, “Assignment Plan,” supplied course document, pp. 1–2. The course plan specifies the Assignment 1.1 deliverables: domain definition, PEAS analysis, environment classification, and a comparison of an autonomous agent with a standard object-oriented program.

[2] Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2021. Used as general background for intelligent-agent and environment terminology.

Submission checklist: Domain/persona defined ✓ PEAS table complete ✓ Environment classification included ✓ Agent-vs-program contrast included ✓